Chapter 13 Vibrations and Waves

Hooke's Law

$$F_S = -kx$$

where F_S is the spring force(restoring force), k is the spring constant, x is the displacement of the object from its equilibrium position. The negative sign indicates that the force is always directed opposite to the displacement.
The energy stored in a stretched or compressed spring or other elastic material is called elastic potential energy.

$$PE_S = \frac{1}{2}kx^2$$

Actually, the spring has mass and oscillates. If the mass of spring are much less than the mass of object, the effective additional mass of a light spring is 1/3 the mass of the spring.

Simple Harmonic Motion

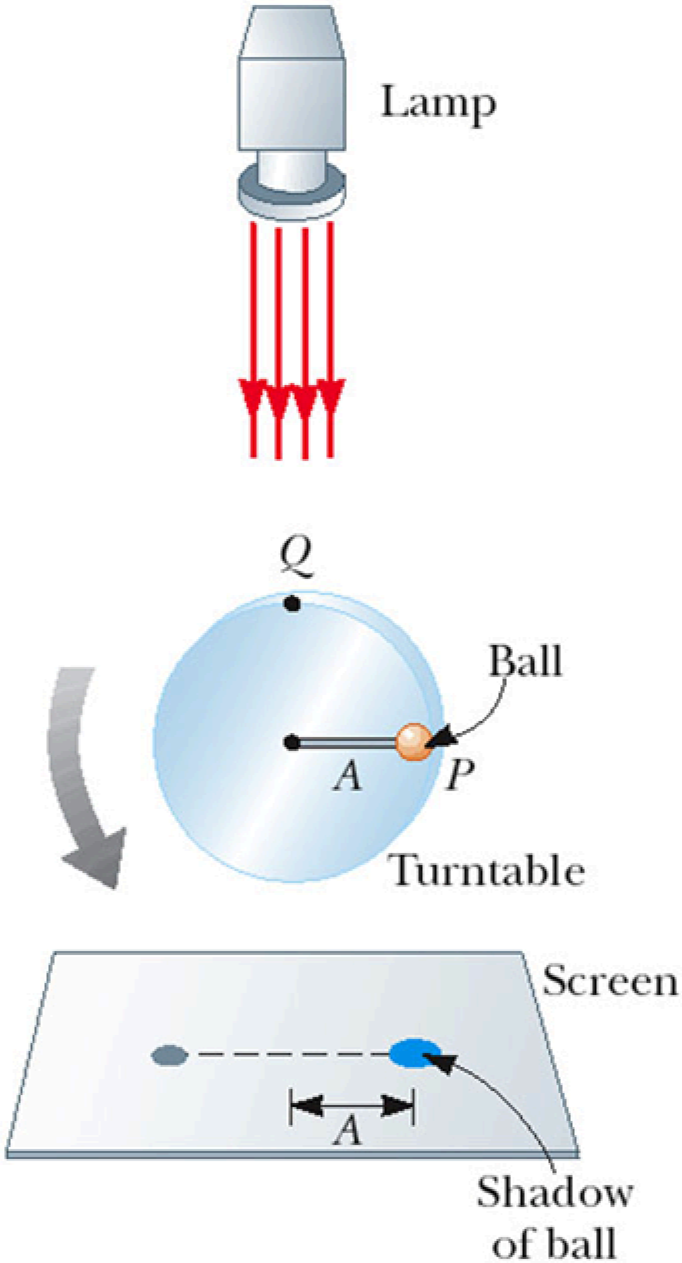
Motion that occurs when the net force along the direction of motion obeys Hooke's Law.

Amplitude is the maximum position of the object relative to the equilibrium position. And the period

$$T = 2\pi\sqrt{\frac{m}{k}}$$

the frequency

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$



If the ball is uniform circular motion, the projectile moves in simple harmonic motion.
And the angular speed

$$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$

The motions function is that

$$x = A\cos(2\pi f t)$$

where x is the position at time t.
If we take derivatives,

$$v = -2\pi f A\sin(2\pi f t) = -A\omega\sin\omega t$$

$$a = -4\pi^2 f^2 A \cos(2\pi f t) = -A\omega^2 \cos \omega t$$

Simple Pendulum

The force is the component of the tangent to the path of motion.

$$F_t = -mg \sin \theta$$

for small angle ($\theta < 5^\circ$), $\sin \theta = \theta$ and $F_t = -mg\theta = -mg\frac{x}{L}$. Which obeys Hooke's Law.

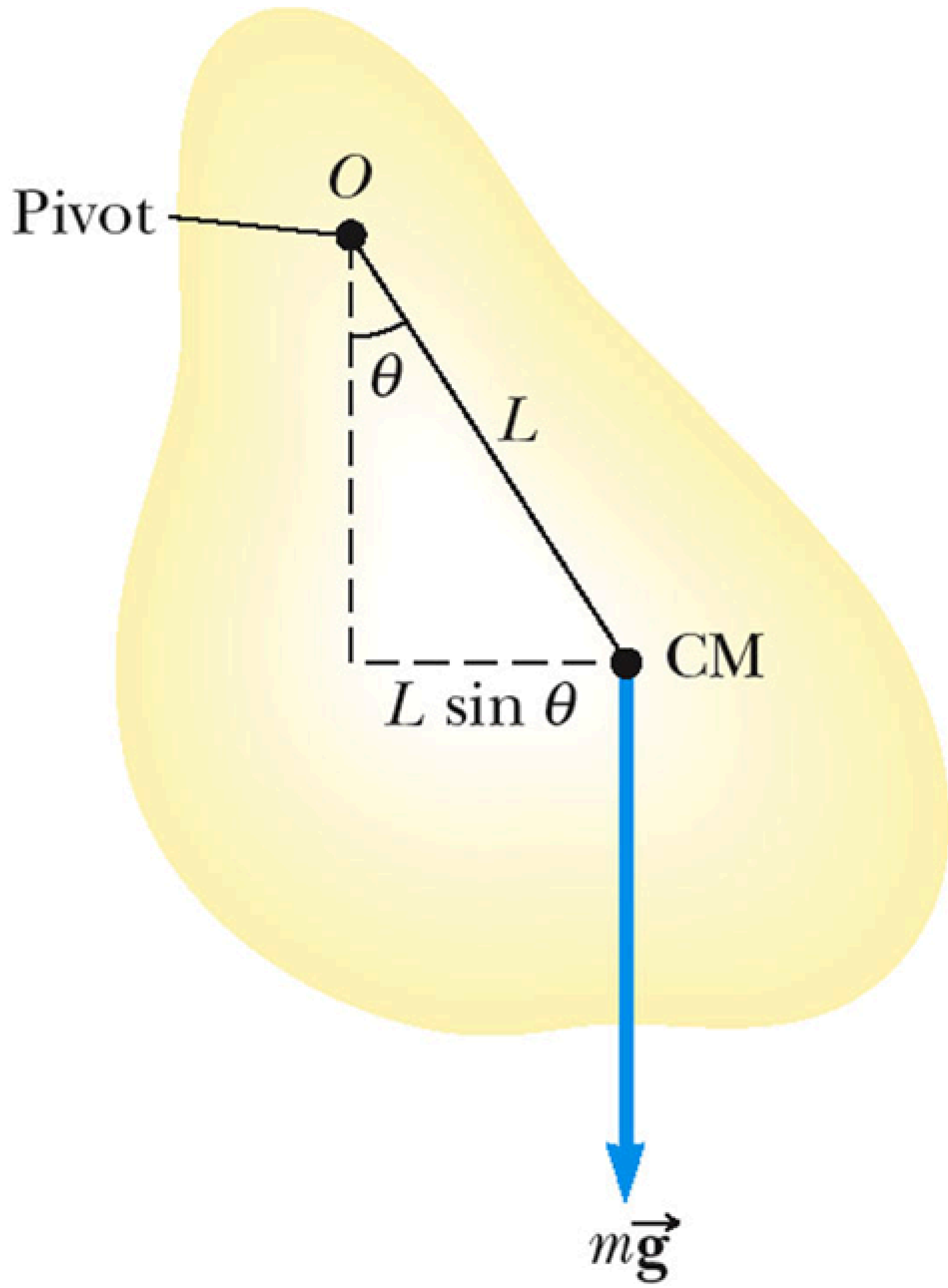
Period is

$$T = 2\pi \sqrt{\frac{L}{g}}$$

For angular simple harmonious motion,

$$\tau = -k\theta$$

Physical Pendulum



The center of mass oscillates along a circular arc.
And the period is

$$T = 2\pi\sqrt{\frac{I}{mgL}}$$

where I is the object's moment of inertia and m is the object's mass. For simple pendulum, $I = mL^2$.

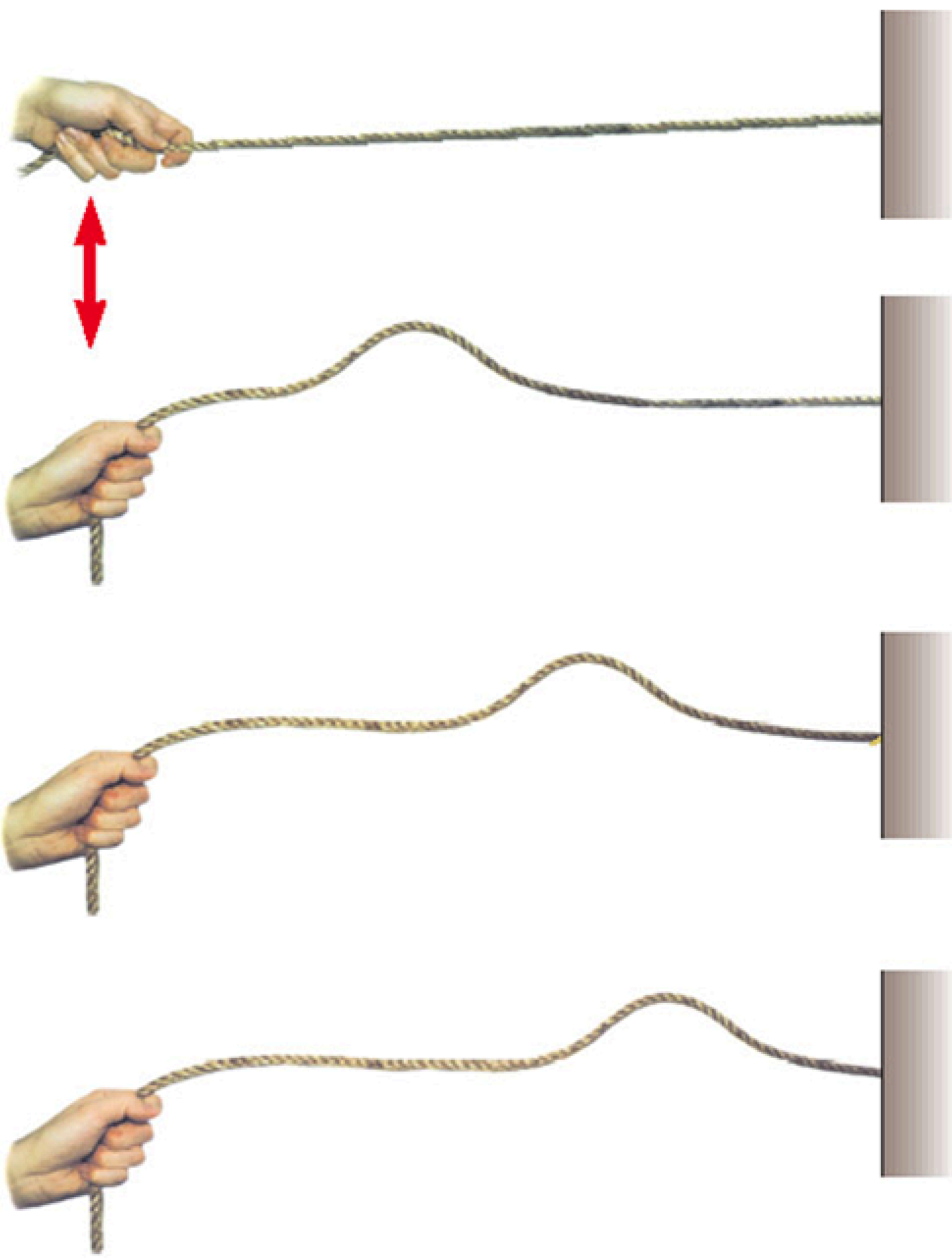
Wave Motion

A wave is the motion of a disturbance, carrying energy and momentum. For mechanical waves, it requires

- Source of disturbance
- Medium that can be disturbed
- Physical connection between or mechanism though which adjacent portions of the medium influence each other.

Types of Waves

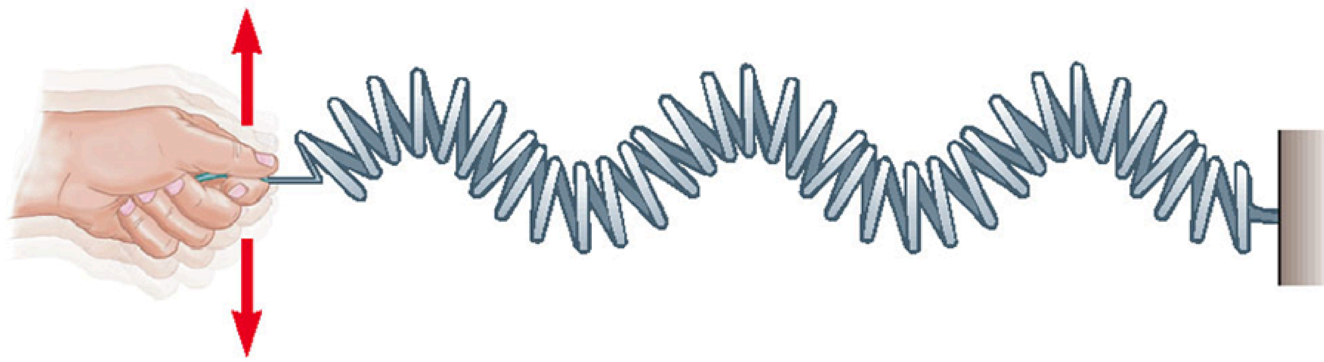
Traveling Waves



One end is fixed and one end is disturbed. The pulse travels to the right with a definite speed. And it is called a traveling wave.

Transverse

Each element that is disturbed moves in a direction perpendicular to the wave motion.

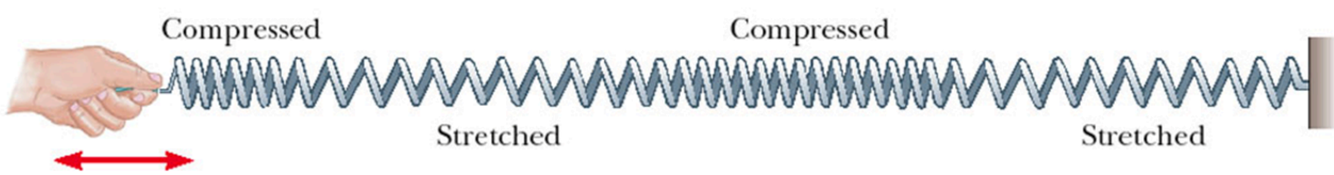


(a) Transverse wave

Longitudinal

The elements of the medium undergo displacements parallel to the motion of the wave.(also called compression wave)

The longitudinal wave can also be represented as a sine curve. Compressions correspond to crests and stretches correspond to troughs.



(b) Longitudinal wave

There exists other types of mechanical waves.

Description of a Wave

For every point in the wave graph, it oscillates in simple harmonic motion in perpendicular direction. Amplitude is the maximum displacement of string above the equilibrium position. Wavelength is the distance between two successive points that behave identically.
And $v = f\lambda$.
For the speed on a wave stretched under some tension F,

$$v = \sqrt{\frac{F}{\mu}}, \mu = \frac{m}{L}$$

where μ is called the linear density.

Interference of Waves

Superposition Principle:
If two or more traveling waves are moving through a medium, the resulting wave is the addition together of all waves at each point. (Only true for waves with small amplitudes)

Constructive Interference

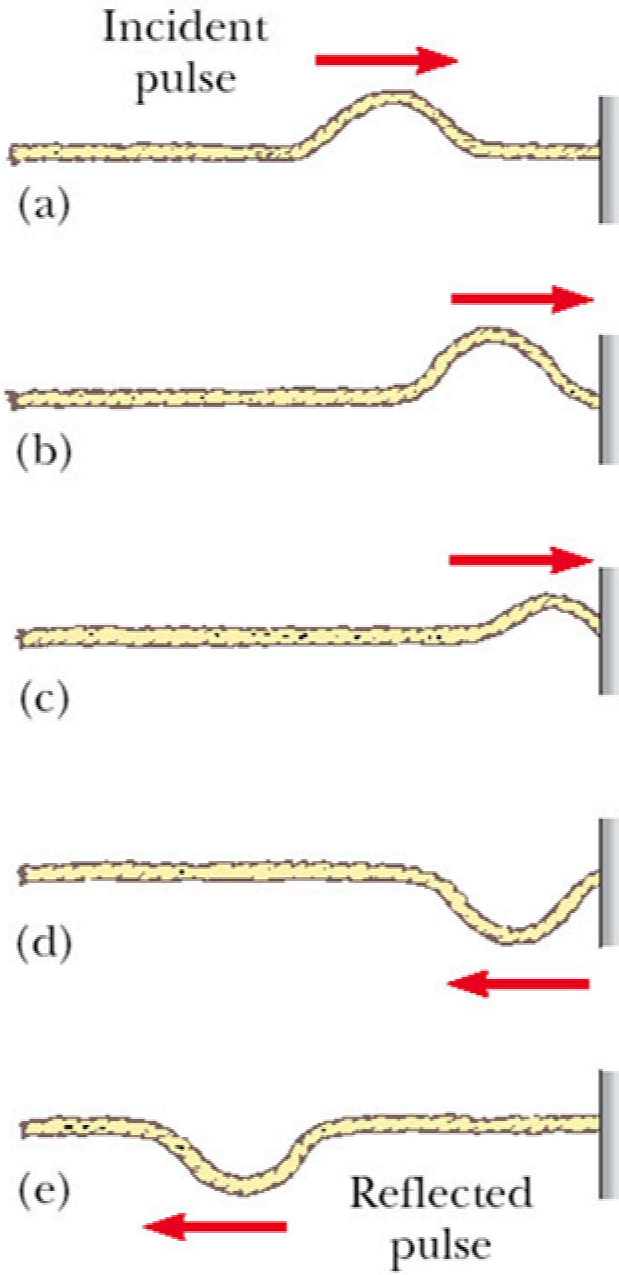
Two waves have the same frequency and are in phase, then the combined wave has the same frequency and a greater amplitude.

Destructive Interference

Two waves have the same frequency and are 180° out of phase, the waveforms are receded.

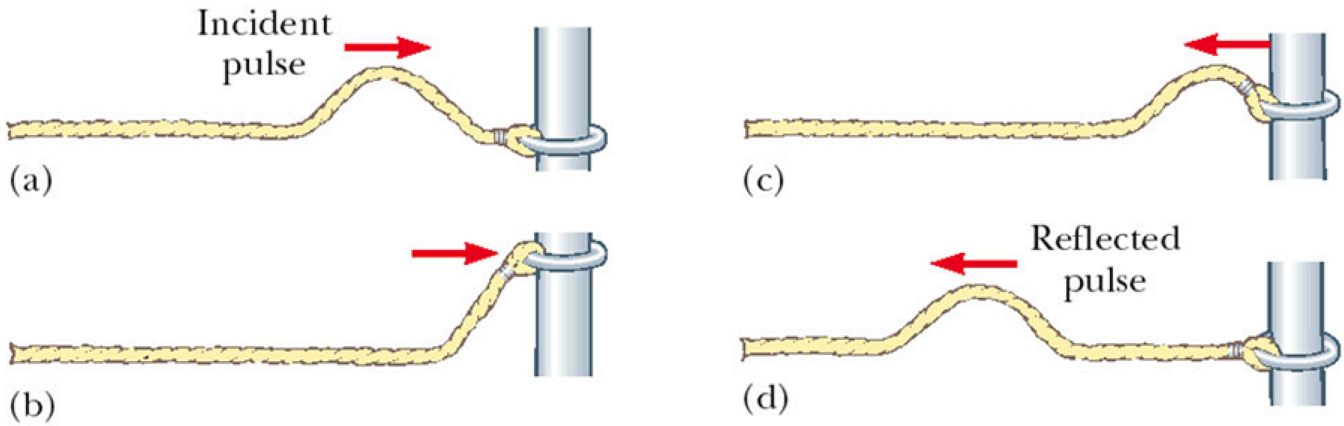
Reflection of Waves

Fixed End



The shape remains the same but the wave is inverted.

Free End



The pulse is not inverted.